

Dr Zhangshu (Joshua) Jiang

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github.com/z0shua | github.com/fastomcp/omcp_py | linkedin.com/in/zhangshu-jiang

PROFILE

LLM evaluation researcher and systems builder, and medical doctor working on post-training, benchmark design, and retrieval-augmented systems for high-stakes domains. PhD research examines long-horizon model behaviour over longitudinal clinical data: QLoRA fine-tuning of open-source models with Unsloth, rubric-based evaluation of factuality and grounding, and multi-step reasoning assessment. Built an open-source natural-language-to-SQL system adopted by five research teams, and currently evaluate frontier model outputs as a Medical Fellow at Scale AI. My clinical background means I notice when an answer sounds right but isn't.

TECHNICAL SKILLS

- **LLM post-training:** Python, PyTorch, Hugging Face Transformers, Unsloth, QLoRA/PEFT, supervised fine-tuning, instruction tuning, vLLM, LangChain
- **Evaluation and quality:** benchmark design, rubric construction, factuality and citation assessment, hallucination evaluation, adversarial testing, failure-mode analysis
- **Search and retrieval:** retrieval-augmented generation, natural-language-to-SQL, advanced SQL, ETL pipelines, knowledge graphs, Neo4j
- **Infrastructure:** Docker, FastAPI, REST APIs, GitHub Actions, Azure/AWS/GCP

EXPERIENCE

PhD Researcher: LLM Evaluation and Applied Generative AI

Oct 2025 – Present

King's College London, EPSRC DRIVE-Health CDT

- Build long-horizon benchmarks for temporal and counterfactual reasoning over longitudinal EHRs. The core problem: plausible intermediate reasoning can still produce a wrong final answer, so the evaluation has to catch that, not just check the outcome.
- Fine-tune Qwen-family models using QLoRA/PEFT with Unsloth across five reasoning task categories, evaluating instruction-following, consistency, and robustness under targeted prompting.
- Built an eight-domain evaluation suite covering factual grounding, citation support, and adversarial robustness across 200+ constructed queries; reduced unsupported outputs by ~30% against a prompted baseline.
- Built a retrieval-augmented generation pipeline over 500+ clinical documents, cutting unsupported recommendations by ~35% versus baseline prompting. Manuscript under review at *Nature Communications*.

Tools: PyTorch, Hugging Face Transformers, Unsloth, QLoRA/PEFT, LangChain

Medical Fellow: Model Evaluation and Rubric Assessment (Part-time)

Jun 2026 – Present

Scale AI

- Evaluate frontier-model outputs in medicine against expert-developed rubrics for factual accuracy, reasoning quality, and safety.
- Contribute to rubric calibration and failure-mode analysis, mostly around spotting answers that sound complete but leave out something that matters clinically.
- Feed recurring error patterns back into the evaluation process to make it more consistent over time.

Clinical AI Research Fellow and Lead ML Architect

Nov 2024 – Oct 2025

Lancashire and South Cumbria Secure Data Environment

OMCP — Open-source LLM Systems Tool

github.com/fastomcp/omcp_py

- Shipped OMCP, an open-source Model Context Protocol server for controlled natural-language access to structured databases, with audit logging, role-based access control, and privilege separation.
- Built a natural-language-to-SQL pipeline with query rewriting, caching, and batching. Sub-second responses across federated data sources spanning four organisations.
- Packaged and documented the system for adoption across five research teams, cutting partner onboarding time by ~60%.
- Processed 2M+ records using transformer-based NLP for entity extraction and normalisation.

Neuroscience Fellow and Digital Health AI Fellow (Part-time)

Sep 2025 – Present

Cleveland Clinic London

- Lead development of AI-assisted decision tools, focusing on interpretability and failure-mode analysis in clinician-facing workflows.
- Translate requirements from 15+ clinical domain experts into data models and evaluation dashboards.

Medical Doctor: ICU, Neurosurgery, Acute Medicine

Aug 2023 – Aug 2025

Lancashire Teaching Hospitals NHS Foundation Trust

- Built and deployed a deterioration-detection alert system across four clinical teams. Cut false-positive rates by

~20% without missing more real events.

- Worked with real-time multimodal ICU data, where ground truth is delayed and the wrong call has real consequences.

OPEN SOURCE AND PUBLICATIONS

- **OMCP:** Lead author and maintainer of an open-source Model Context Protocol server for auditable LLM access to structured databases via natural-language-to-SQL. github.com/fastomcp/omcp_py
- **Jiang et al.** “Grounding Large Language Models in Clinical Evidence: A Retrieval-Augmented Generation System for Querying UK NICE Clinical Guidelines.” *Under review, Nature Communications.*

SELECTED PROJECTS

- **Dyad Labs (concept and design phase):** AI-native simulator for evaluating human–AI performance in clinical workflows, comparing clinician-only, model-only, and clinician-plus-model outcomes. Developed as a pitch for Perplexity’s Billion Pound Build.
- **62M-node Biomedical Knowledge Graph:** Retrieval system over 12 ontologies. Optimised traversal cut query latency by 65%.
- **Federated Analytics Platform:** Privacy-preserving analytics across three NHS trusts and 500K+ patient records, cutting cross-trust query time by ~50%.

EDUCATION

MPhil/PhD, Health Data Science and Medical AI

2025 – 2029 expected

King’s College London, EPSRC DRIVE-Health CDT

Focus: LLM evaluation, post-training, long-horizon benchmarks, counterfactual reasoning, retrieval-augmented systems

MBChB, Medicine

2018 – 2023

University of Sheffield

GMC full registration with licence to practise

ADDITIONAL

- Delivered AI and data-science workshops to 100+ students and clinicians across four UK regions.
- Languages: English (native), Mandarin Chinese (native), Spanish (professional).